Psychiatric Quarterly (2026) 97:865–878
https://doi.org/10.1007/s11126-025-10185-z

ORIGINAL PAPER

Peripheral Inflammation In Adult ADHD: Sex Differences And Hematological Markers

Pınar Kızılay Çankaya[1,2]

Accepted: 29 June 2025 / Published online: 10 July 2025
© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2025

Abstract

Attention-Deficit/Hyperactivity Disorder (ADHD) is a neurodevelopmental disorder characterized by impairments that persist into adulthood. Despite advancements in our understanding of ADHD, the factors influencing its course remain unclear. The inflammatory hypothesis suggests that immune dysregulation may contribute to ADHD pathophysiology. However, research examining inflammation in adult ADHD is limited. This cross-sectional study aimed to compare complete blood count (CBC) parameters and derived inflammatory indices between adult patients with ADHD and healthy controls, with particular attention to sex differences. The study included 72 drug-free adults with ADHD (aged 18–38 years) diagnosed according to the Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5) and 72 age- and sex-matched healthy controls. CBC parameters and inflammatory indices, including neutrophil-to-lymphocyte ratio (NLR), platelet-to-lymphocyte ratio (PLR), monocyte-to-lymphocyte ratio (MLR), basophil-to-lymphocyte ratio (BLR), mean platelet volume (MPV), platelet distribution width (PDW), systemic immune-inflammatory index (SII), and pan-immune inflammatory value (PIV), were compared between the groups. Monocyte counts were significantly higher in the ADHD group than in the control group ($p = 0.019$). Sex-specific analysis revealed no significant differences in inflammatory parameters among females. However, male patients with ADHD showed significantly elevated leukocyte, neutrophil, and monocyte counts compared to male controls ($p = 0.031$, $p = 0.049$, and $p = 0.038$, respectively). These findings suggest that elevated monocyte levels, indicative of chronic inflammation, may have diagnostic implications in adult ADHD, particularly among males. These findings underscore the necessity to incorporate sex-specific biological mechanisms when investigating and developing therapeutic strategies for ADHD.

Keywords ADHD · Inflammation · Monocytes · Sex differences · Immune dysregulation · Hematological markers

✉ Pınar Kızılay Çankaya
pkizilay@gmail.com

[1] Private Practice, Psychiatry, Ankara, Türkiye

[2] Institute of Health Sciences, Department of Medical Pharmacology, Gazi University, Ankara, Türkiye

Introduction

Attention-Deficit/Hyperactivity Disorder (ADHD) is a common and multifaceted neurodevelopmental condition that manifests as patterns of inattention, excessive activity, and reduced impulse control, substantially impairing academic, social, and professional functioning [1, 2]. Beyond its core symptoms, ADHD commonly presents alongside other psychiatric conditions, including depressive disorders, anxiety, and substance use disorders, further exacerbating functional impairments and diminishing quality of life [1, 3]. Although historically conceptualized as a childhood disorder, longitudinal studies have confirmed its persistence into adulthood in 50–70% of cases [4, 5]. As ADHD symptoms evolve throughout life, the complex interplay between genetic and environmental factors is believed to contribute to the persistence of the disorder [4]. The mechanisms underlying sex differences in various developmental processes in ADHD patients remain poorly understood [6].

Despite extensive research, the precise etiology of ADHD remains elusive, involving complex interactions between genetic, neurobiological, and environmental factors [4, 7–11]. However, emerging evidence suggests that immune system dysregulation and low-grade chronic inflammation may significantly contribute to pathophysiology [12–16]. This hypothesis aligns with broader findings implicating neuroinflammatory processes in other neuropsychiatric disorders, including depression, schizophrenia, and autism spectrum disorder [17–19]. In recent years, research has increasingly focused on investigating how systemic inflammation and immune dysregulation contribute to ADHD [20, 21]. Studies have indicated that conditions with inflammatory components, such as eczema, asthma, rheumatoid arthritis, type 1 diabetes, and hypothyroidism, are more common in individuals with ADHD [22, 23]. It remains unclear whether inflammation plays a causal role in the persistence of ADHD symptoms into adulthood or whether immune activation occurs as a secondary consequence of ADHD-related stressors and comorbidities [11].

Sex-based differences in immune responses represent an important yet underexplored area in ADHD research that might help explain these sex differences. This connection is particularly relevant because most autoimmune diseases are more prevalent in women, and the incidence and clinical presentation of other immune-related conditions vary between sexes [24]. These variations can be attributed to the crucial role of sex hormones in regulating immune responses, with testosterone generally having immunosuppressive effects and estrogen typically enhancing immune activity [25]. Consistent with this pattern, adult females of many species mount more robust antibody responses to immune challenges than males [26].

Throughout childhood and adolescence, the immune system changes, partly due to fluctuating levels of steroids and sex hormones during these developmental stages [27, 28]. This makes it challenging to compare the immune signatures from childhood or puberty to those of adult patients. Sex hormones, particularly estrogen, progesterone, and androgens, play crucial roles in modulating immune cell function and responses [26, 29, 30]. Environmental factors such as chronic stress, dietary habits, and physical activity also influence immune responses and may interact with sex-related hormonal mechanisms [20]. The question of sex-based differences in the immune response in ADHD remains underexplored, despite well-documented differences in both clinical presentation and immune regulation between males and females [31, 32].

Blood-based inflammatory markers derived from complete blood count (CBC) provide a cost-effective and accessible method for assessing systemic inflammation [33, 34].

Several ratios and indices calculated from CBC parameters, including the neutrophil-to-lymphocyte ratio (NLR), platelet-to-lymphocyte ratio (PLR), monocyte-to-lymphocyte ratio (MLR), basophil-to-lymphocyte ratio (BLR), mean platelet volume (MPV), and platelet distribution width (PDW), have been used to evaluate inflammation in various neuropsychiatric disorders [35–38]. The Systemic Immune-Inflammation Index (SII) offers a more integrated representation of immune system activity [39]. The Pan-Immune Inflammation Value (PIV) is elevated in adults with obsessive–compulsive disorder (OCD) and children with ADHD [40, 41].

Studies examining the role of inflammation in ADHD have yielded conflicting results, largely due to small sample sizes, heterogeneous study populations, and variations in the inflammatory markers examined [42–48]. Understanding this relationship holds promise in elucidating the etiology of this disorder and identifying novel therapeutic targets. Despite the increasing recognition of adult ADHD, only one study to date has investigated peripheral inflammation indicators in adult patients, finding no significant differences compared to controls and no association with treatment or clinical symptoms [47, 49]. Notably, to the best of our knowledge, no studies have evaluated peripheral inflammation indicators concerning sex differences in adult patients with ADHD.

The present study addresses this gap by comparing CBC-derived inflammatory markers, including NLR, PLR, MLR, BLR, MPV, PDW, SII, and PIV, between drug-free adult patients with ADHD and healthy controls. We also investigated the sex-based differences in these parameters. Clarifying the role of peripheral inflammation and sex-specific immune responses may support the development of targeted and individualized treatment strategies for adult ADHD.

Methods

Ethical Approval

Ethical approval was obtained from the Ethics Committee of Koru Ankara Hospital on. September 12, 2023 (approval number: 3201).

Sample

We recruited 144 participants from the psychiatric outpatient clinic of Koru Ankara Hospital between April 2022 and April 2023. The sample comprised 72 drug-free adults (aged 18–38 years) diagnosed with ADHD according to the Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5) criteria and 72 age- and sex-matched healthy controls. The control participants were aviation personnel who had undergone health examinations at the Koru Hospital Aeromedical Center and were certified to be in optimal health. A power analysis using G*Power (version 3.1.9.7) indicated that a minimum of 34 participants per group would achieve 80% power to detect a large effect size (0.70) at a significance level of $\alpha = 0.050$ using a two-sided two-sample equal-variance t-test. Data were collected retrospectively through a medical record review by the author affiliated with the outpatient clinic.

This cross-sectional study initially enrolled 125 patients with ADHD; however, after applying the exclusion criteria, the final sample consisted of 72 adult patients with ADHD and 72 healthy controls, all of whom had CBC values within the normal reference ranges.

Participants with abnormal hematological values, including anemia, polycythemia, lymphopenia, lymphocytosis, neutropenia, neutrophilia, thrombocytopenia, or thrombocytosis, were excluded. Sociodemographic data of all participants were retrospectively obtained from their medical records.

The exclusion criteria for the ADHD group included psychiatric comorbidities, current medication use, or any medical conditions potentially influencing inflammatory parameters. Specifically, we excluded individuals with inflammatory or autoimmune diseases, hematological disorders, liver or renal failure, clinical evidence of active infections, obesity, heavy smoking (defined as smoking > 15 cigarettes per day), or use of anti-inflammatory or immunosuppressive medications. Additionally, individuals whose ADHD diagnosis was revised during follow-up or whose medical records contained incomplete or conflicting data were excluded.

All laboratory analyses were conducted at the Koru Ankara Hospital Medical Biochemistry and Medical Microbiology Laboratory using an automated hematological analyzer (Sysmex 1800i). Blood samples were collected as part of the participants'routine laboratory evaluation during their clinical visit, typically between 8:00 and 10:00 a.m. However, the exact time of day for blood collection was not standardized across participants, as the data were collected retrospectively. The study adhered to the standard reference intervals for hematological parameters as follows: leukocytes ($4.8–10.8 \times 10^3/mm^3$), neutrophils ($2.06–7.02 \times 10^3/mm^3$), lymphocytes ($1.3–3.5 \times 10^3/mm^3$), platelets ($150–450 \times 10^3/mm^3$), monocytes ($0.3–0.8 \times 10^3/mm^3$), eosinophils ($< 0.42 \times 10^3/mm^3$), and basophils ($< 0.2 \times 10^3/mm^3$). An automated hematology analyzer was used to measure the PDW and MPV. SII was calculated using the following formula: SII = (platelet count × neutrophil count)/lymphocyte count. PIV was calculated using the following formula: PIV = (platelet count × neutrophil count × monocyte count)/lymphocyte count. NLR, PLR, MLR, and BLR were calculated by dividing the absolute counts of the relevant cells.

Statistical Analysis

Statistical analyses were performed using IBM Statistical Package for Social Sciences (SPSS) version 22. Categorical variables were presented as frequencies and percentages. Continuous variable normality was evaluated through visual assessment (histograms and probability plots) and analytical methods (Kolmogorov–Smirnov and Shapiro–Wilk tests). For descriptive statistics, normally distributed variables were presented as means with standard deviations, while non-normally distributed variables were expressed as medians with interquartile ranges (1st and 3rd quartiles).

Between-group comparisons of continuous variables were performed using Student's t-test or the Mann–Whitney U test according to the distribution characteristics. Chi-square analyses were performed for categorical variables, applying either Pearson's chi-square or Fisher's exact test, as methodologically appropriate. Statistical significance was established at $p < 0.05$ for all analyses.

Results

This study included 72 participants diagnosed with ADHD and 72 healthy controls. The ADHD group consisted of 36 females (50%) and 36 males (50%), while the control group included 39 females (54.2%) and 33 males (45.8%). The median age was 19 years (range:

18–21 years) in the ADHD group and 20 years (range: 19–23 years) in the control group. Statistical analysis revealed no significant differences between the groups in terms of age ($p = 0.090$) or sex distribution ($p = 0.617$) (Table 1).

Table 1 presents a comparison of hematological parameters between the ADHD and control groups. No statistically significant differences were observed in NLR, PLR, MLR, BLR, SII, or PIV ($p > 0.05$). Similarly, platelet indices, including mean platelet volume (MPV) and platelet distribution width (PDW), did not differ significantly between the groups ($p > 0.05$). However, analysis of the CBC results revealed significantly higher monocyte counts in the ADHD group than in the control group ($z = -2.347$, $p = 0.019$) (Table 1). The effect size for the group difference in monocyte counts was large, with Cohen's d of 1.66. The rank-biserial correlation for this comparison was –0.76, indicating a strong association between ADHD diagnosis and elevated monocyte levels.

Sex-stratified analyses revealed different patterns between males and females. Among female participants, no statistically significant differences were observed in any hematological parameters between the ADHD and control groups ($p > 0.05$) (Table 2). In contrast, the male participants showed significant differences in several parameters. Males with ADHD demonstrated higher leukocyte counts ($t = -2.200$, $p = 0.031$) and significantly elevated neutrophil counts ($t = -2.006$, $p = 0.049$) than the male controls. Additionally, monocyte counts were significantly higher in males with ADHD than in male controls ($z = -2.079$, $p = 0.038$) (Table 3). Among males, the effect size for the difference in monocyte counts

Table 1 Age, sex, and hematological parameters in ADHD and control groups

	Patients with ADHD n = 72	Healthy controls n = 72	x^2/z/t	*p*
Age (years)	21.00 (19.00–25.75)	23.00 (20.00–25.75)	−1.696	0.090*
Sex			0.250	0.617**
Male	36 (50%)	33 (45.8%)		
Female	36 (50%)	39 (54.2%)		
Leukoyctes $10^3/mm^3$	7.15 ± 1.80	6.74 ± 1.45	−1.494	0.137***
Neutrophils $10^3/mm^3$	4.03 ± 1.27	3.81 ± 0.96	−1.178	0.241***
Lymphocytes $10^3/mm^3$	2.36 ± 0.78	2.26 ± 0.54	−0.940	0.349***
Monocytes $10^3/mm^3$	0.51 (0.43–0.63)	0.45 (0.37–0.59)	**−2.347**	**0.019***
Basophils $10^3/mm^3$	0.04 (0.03–0.06)	0.04 (0.03–0.05)	−0.278	0.781*
Eosinophils $10^3/mm^3$	0.12 (0.07–0.21)	0.11 (0.07–0.19)	−0358	0.720*
Platelets $10^3/mm^3$	264.04 ± 58.15	264.53 ± 57.12	0.051	0.960***
MPV fL	9.90 (9.40–10.50)	9.90 (9.30–10.40)	−0.604	0.546*
PDW %	11.40 (10.10–12.95)	10.80 (9.90–11.90)	−1.415	0.157*
NLR	1.62 (1.31–2.18)	1.68 (1.39–2.01)	−0.274	0.784*
PLR	133.71 ± 136.85	121.59 ± 30.74	−0.733	0.465***
BLR	0.02 (0.01–0.03)	0.02 (0.01–0.02)	−0.278	0.781*
MLR	0.22 (0.19–0.22)	0.22 (0.17–0.26)	−1.394	0.163*
SII	440.39 (325.32–588.75)	450.75 (349.64–526.64)	−0.092	0.927*
PIV	221.57 (161.16–329.52)	205.91 (149.32–314.27)	−1.131	0.258*

*Mann–Whitney U test used for analysis; **Chi-square test used for analysis; ***Two-sample *t-test* used for analysis; p values less than 0.05 were shown in bold

Table 2 Hematological parameters in female ADHD and control groups

	Female patients with ADHD n = 36	Female healthy controls n = 39	x^2/z/t	*p*
Leukoyctes $10^3/mm^3$	6.77 ± 1.58	6.83 ± 1.55	0.170	0.866***
Neutrophils $10^3/mm^3$	3.80 ± 1.08	3.90 ± 1.01	0.443	0.659***
Lymphocytes $10^3/mm^3$	2.39 (1.82—2.66)	2.20 (1.75—2.68)	−0.095	0.924*
Monocytes $10^3/mm^3$	0.49 (0.40—0.57)	0.43 (0.37—0.53)	−1.040	0.298*
Basophils $10^3/mm^3$	0.04 (0.03–0.05)	0.04 (0.03–0.05)	−0.971	0.332*
Eosinophils $10^3/mm^3$	0.10 (0.06—0.19)	0.10 (0.07—0.16)	−0.292	0.770*
Platelets $10^3/mm^3$	264.39 ± 67.54	274.46 ± 48.57	0.746	0.458***
MPV fL	10.18 ± 0.82	10.02 ± 0.88	−0.604	0.546*
PDW %	11.45 (10.28—14.20)	10.80 (9.90–11.90)	−1.483	0.138*
NLR	1.77 ± 0.64	1.77 ± 0.46	0.003	0.998***
PLR	122.94 ± 40.69	126.27 ± 30.12	0.404	0.687***
BLR	0.01 (0.01—0.03)	0.02 (0.01—0.02)	−0.811	0.417*
MLR	0.23 ± 0.07	0.21 ± 0.05	−1.412	0.163***
SII	472.93 ± 218.44	486.99 ± 154.08	0.324	0.747***
PIV	209.37 (140.83—307.88)	203.44 (153.20—316.32)	−0.021	0.983*

*Mann–Whitney U test used for analysis; **Chi-square test used for analysis; ***Two-sample *t-test* used for analysis; p values less than 0.05 were shown in bold

Table 3 Hematological parameters in male ADHD and control groups

	Male patients with ADHD n = 36	Male healthy controls n = 33	x^2/z/t	*p*
Leukoyctes $10^3/mm^3$	7.53 ± 1.95	6.63 ± 1.34	**−2.200**	**0.031*** **
Neutrophils $10^3/mm^3$	4.26 ± 1.42	3.69 ± 0.89	**−2.006**	**0.049*** **
Lymphocytes $10^3/mm^3$	2.43 ± 0.86	2.24 ± 0.52	−1.105	0.273***
Monocytes $10^3/mm^3$	0.54 (0.47—0.68)	0.47 (0.40—0.60)	**−2.079**	**0.038***
Basophils $10^3/mm^3$	0.04 (0.03–0.06)	0.04 (0.03–0.05)	−1.361	0.173*
Eosinophils $10^3/mm^3$	0.15 (0.09—0.24)	0.15 (0.08—0.21)	−0.511	0.609*
Platelets $10^3/mm^3$	263.69 ± 47.94	252.79 ± 64.62	−0.801	0.426***
MPV fL	9.90 (9.18–10.38)	9.80 (9.15–10.25)	−0.361	0.718*
PDW %	11.22 ± 1.70	11.07 ± 7.71	−0.370	0.713***
NLR	1.59 (1.32—2.16)	1.64 (1.36—2.00)	−0.006	0.995*
PLR	144.48 ± 189.99	116.07 ± 30.99	−0.848	0.399***
BLR	0.02 (0.01—0.03)	0.02 (0.01—0.02)	−0.511	0.610*
MLR	0.22 (0.19—0.32)	0.24 (0.18–0.27)	−0.697	0.486*
SII	424.77 (339.70–519.66)	419.86 (304.18–515.83)	0.565	0.572*
PIV	225.00 (168.39–342.55)	223.83 (115.70–311.23)	−1.309	0.190*

*Mann–Whitney U test used for analysis; **Chi-square test used for analysis; ***Two-sample *t-test* used for analysis; p values less than 0.05 were shown in bold

was also substantial, with a Cohen's d of 1.66 and a rank-biserial correlation of –0.76, further underscoring the clinical relevance of this finding. For leukocyte count, the effect size was large (Cohen's d = 3.35; rank-biserial correlation = –1.00), suggesting a strong relationship between ADHD and elevated leukocyte levels in males. For the neutrophil count, a similarly large effect was observed (Cohen's d = 3.42; rank-biserial correlation = –1.00), supporting the robustness of the group difference. Figure 1 shows a boxplot of sex-specific differences in leukocyte, neutrophil, and monocyte counts.

Discussion

This study explored the link between peripheral inflammation and adult ADHD, with a specific focus on sex differences. The results indicated that while most inflammatory markers, such as NLR, BLR, MLR, PLR, SII, PIV, and platelet indices, showed no significant differences between the ADHD and control groups, monocyte counts were significantly higher in the ADHD group. Sex-stratified analyses revealed that these differences were driven by the male sex, with male ADHD participants exhibiting higher leukocyte, neutrophil, and monocyte counts than male controls, whereas no such differences were observed in females.

These findings provide meaningful insight into the role of the immune system in ADHD. The elevated monocyte counts suggest that chronic low-grade inflammation might play a role in the disorder, particularly in males. Monocytes are key players in the innate immune response and contribute to neuroinflammation [50]. Previous research has linked increased monocyte levels to other neuropsychiatric conditions such as schizophrenia and depression, suggesting that immune dysregulation may be a broader feature across multiple psychiatric disorders [51, 52]. The clinical utility of elevated monocyte levels in adult ADHD remains a subject of future research. Monocyte elevation in ADHD, particularly among males, may reflect the underlying neuroinflammatory processes. However, given the modest effect size and lack of elevation in other inflammatory indices in our study, monocyte count alone may lack sufficient specificity or sensitivity to function as a standalone biomarker. Instead, it could potentially serve as part of a broader panel of hematological and molecular indicators to identify inflammatory subtypes of ADHD. Such stratification may assist in predicting treatment response or tailoring adjunctive anti-inflammatory therapies. Future studies should examine the temporal stability, reproducibility, and diagnostic utility of monocyte elevations across larger and more diverse ADHD populations,

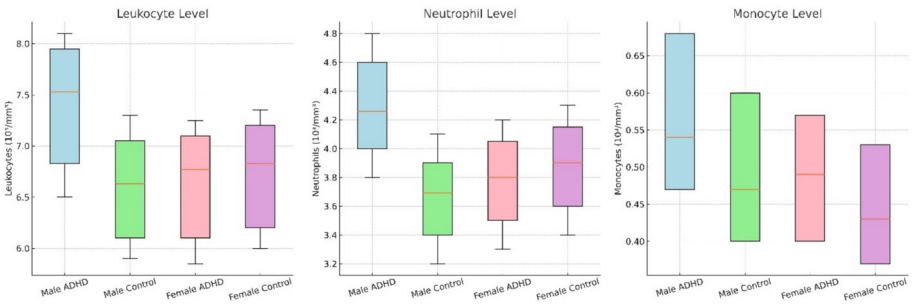


Fig. 1 Boxplot for sex-stratified analyses

ideally incorporating functional and molecular immune assays to validate their biomarker potential.

Sex-specific differences in inflammation represent a key contribution to this study. Elevated leukocyte, neutrophil, and monocyte counts were observed in male participants with ADHD, whereas no such changes were observed in the female participants. These results support previous evidence suggesting sex-based neurobiological and immunological variations in ADHD [53]. This aligns with existing literature indicating that males generally exhibit stronger pro-inflammatory profiles than females, potentially owing to the immunosuppressive effects of testosterone [26]. In contrast, estrogen enhances immune reactivity, which may explain the stronger immune responses observed in females during childhood but not necessarily during adulthood [26, 42]. Akıncı and Uzun (2021) found higher neutrophil counts and NLR values in female children with ADHD, suggesting stronger acute immune responses in female pediatric populations [42]. Our study identified a reversed pattern in adults, with higher inflammatory indicators in males. These findings imply that sex differences in inflammatory mechanisms may evolve through hormonal and environmental modulation [54]. Our findings emphasize that inflammation may play a more prominent role in adult male ADHD, contributing to sex-specific expression of the disorder.

However, some studies reported no significant differences in peripheral inflammation markers between patients with ADHD and controls, underscoring the inconsistency in the literature on ADHD [47]. These discrepancies may arise from methodological differences such as study design, sample characteristics, and the specific inflammatory markers assessed. Additionally, the immune response dynamics may differ across developmental stages. In pediatric ADHD populations, immune responses are often more reactive or acute, whereas in adults, the immune system tends to exhibit a more stable but low-grade, chronic inflammatory profile [14, 55].

Compared to studies in pediatric populations, our findings suggest a potentially distinct immunological profile in adults with ADHD. While several studies in children and adolescents have reported elevated inflammatory indices such as NLR and PLR, particularly in females [42, 55, 56], our adult sample did not demonstrate these elevations. Instead, we observed elevated monocyte, neutrophil, and leukocyte counts specifically in males, indicating a sex-specific chronic low-grade inflammatory response in adulthood. This discrepancy may reflect developmental transitions in immune function, age-related hormonal changes, or cumulative environmental stressors that shape immune responses differently over time [26, 27]. The elevated immune markers observed in our sample align with previous findings that highlight the role of long-term stress in shaping immune responses [57]. These findings support the moderating role of sex in the pathophysiology of ADHD and highlight the need for sex-specific treatment approaches. To our knowledge, this is the first study to report sex-differentiated CBC-based inflammatory patterns in drug-free adults with ADHD, contributing new insights into how immune dysregulation may evolve beyond childhood. These findings highlight the need to consider age and sex as critical dimensions in ADHD biomarker research and support calls for longitudinal studies linking pediatric inflammatory profiles with adult outcomes.

One possible explanation for the absence of significant inflammatory differences among females with ADHD is related to the immunomodulatory effects of estrogen. Estrogen is known to exert broad anti-inflammatory effects by modulating cytokine production, enhancing regulatory T-cell function, and suppressing pro-inflammatory pathways, such as NF-κB signaling [26, 28, 30]. These mechanisms may attenuate peripheral inflammatory responses in adult females and account for the muted immune signature observed in our female subgroup. In contrast, testosterone, which generally has immunosuppressive effects,

may be less effective in suppressing chronic low-grade inflammation in males, particularly under persistent stress [25, 54]. Furthermore, developmental timing and cumulative lifetime exposure to sex hormones may shape immune system sensitivity differently in men and women [27, 29]. Methodologically, our sex-stratified analyses may also have been underpowered to detect smaller effects in females, necessitating replication in larger samples. Future studies should incorporate hormonal profiling and sex-specific immunophenotyping to clarify the biological underpinnings of these divergent patterns [31, 32, 53].

In contrast to monocyte findings, our study did not find significant differences in other commonly used inflammatory markers, such as NLR, PLR, or SII, between patients with ADHD and controls. These findings are consistent with those of Ceyhun and Gürbüzer (2022), who also found no significant variation in these markers among adults with ADHD [47]. However, several pediatric studies have demonstrated elevated NLR and PLR in children with ADHD, especially in females [42]. These inconsistencies likely reflect developmental changes in immune function and the influence of contextual factors, such as chronic stress, lifestyle, and comorbidities, all of which may shape inflammatory patterns differently in adults and children [20].

Our study revealed no significant elevation in platelet count, MPV, or PDW values among patients with ADHD compared to controls. This finding is consistent with a study of patients with ADHD, which similarly found no differences in MPV and PLT values between adults with ADHD and controls [47]. Interestingly, children with ADHD have previously demonstrated significantly higher MPV values than healthy controls [56]. While these indices have been linked to increased platelet activation in certain neuropsychiatric conditions, particularly depression [58, 59], the stability of these parameters in our adult ADHD sample may indicate that the inflammatory profile of this disorder is more closely associated with immune cells such as monocytes and neutrophils than platelets [47].

Despite these findings, our study did not assess inflammatory differences across ADHD subtypes (i.e., inattentive, hyperactive-impulsive, or combined). The existing literature suggests that inflammatory profiles may vary according to the subtype, although the evidence is mixed [49]. Several studies have reported elevated pro-inflammatory cytokine levels in children with hyperactive-impulsive and combined subtypes, whereas Binici and Kutlu (2018) found no subtype-related differences in the NLR or PLR [44, 55]. Similarly, Misiak et al. (2022) reported uniform inflammatory patterns across ADHD subtypes in adults [52]. These conflicting results may reflect heterogeneity within and between subtypes, with methodological variability.

Beyond its biological implications, these findings raise hypotheses of potential clinical relevance. If future studies establish a consistent and causal relationship between inflammation and ADHD, it may be possible to explore anti-inflammatory interventions as adjunctive treatment strategies. While some preliminary investigations have examined the effects of NSAIDs and dietary components such as omega-3 fatty acids on ADHD symptoms, these results remain inconclusive and insufficient to inform clinical practice at this stage [49, 60, 61]. Therefore, the current implications should be considered exploratory, and further research is needed to determine whether modulation of inflammatory pathways can meaningfully impact ADHD outcomes [49, 62].

Limitations

This study has several limitations. First, its cross-sectional design limits causal inferences. Longitudinal studies are required to clarify whether inflammation contributes to the onset

of ADHD or is a consequence of chronic symptoms and stress. Second, although our a priori power analysis was designed for overall group comparisons, the study was not specifically powered to detect sex-stratified effects. Each sex subgroup consisted of approximately 36 participants, which limits the statistical power to detect smaller effect sizes in within-sex comparisons. Thus, the sex-specific findings should be interpreted with caution. We recommend that future studies recruit larger, sex-balanced cohorts to validate and extend our observations. Third, we did not assess several important confounding variables known to influence the inflammatory status, including body mass index (BMI), sleep quality, dietary habits, physical activity levels, and chronic stress exposure. These lifestyle and physiological factors can significantly modulate immune function and may have contributed to the variability in inflammatory markers independent of ADHD diagnosis. Although efforts have been made to exclude individuals with obesity or clinically evident inflammatory conditions, the absence of standardized assessments for these factors limits the interpretability of our findings. Future studies should incorporate validated measures of metabolic, behavioral, and psychosocial variables to better isolate ADHD-specific immune alterations and improve causal inference. Fourth, the use of peripheral blood markers may not fully capture neuroinflammatory processes. Fifth, the retrospective nature of data collection may introduce bias owing to incomplete clinical documentation, variability in diagnostic procedures, and lack of standardization in the timing of measurements. Furthermore, because the timing of blood collection was not standardized, circadian variations in inflammatory markers may have introduced variability into the results. Future studies should control for the timing of the sample collection to reduce confounding factors. Future studies using prospective designs, advanced neuroimaging, and broader biomarker panels may elucidate the neuroimmune mechanisms underlying ADHD.

Another limitation is the use of aviation personnel as a control group. Although these individuals were medically certified as healthy and age- and sex-matched to the ADHD group, their occupational context may introduce systematic differences. Aviation professionals are subject to rigorous physical and psychological health standards and often exhibit lifestyle patterns that differ from those of the general population. These factors may confound inflammatory parameters and reduce the ecological validity of comparisons with the ADHD group. Therefore, caution is warranted when generalizing these findings to broader and more heterogeneous populations.

An additional consideration is the exclusion of participants with psychiatric comorbidities. Although this approach aimed to isolate ADHD-specific effects, it may have limited the generalizability of the findings. Given that ADHD commonly co-occurs with anxiety, depression, and substance use disorders, each of which is associated with immune dysregulation, future research should differentiate inflammation specifically related to ADHD from that associated with comorbid conditions. Furthermore, the influence of genetic and epigenetic factors on immune function in ADHD warrants investigation to clarify individual differences in disease progression and treatment response [13].

Conclusion

In conclusion, this study provides preliminary evidence suggesting a potential role for immune system involvement in adult ADHD, particularly in males. While most inflammatory markers did not differ significantly between the groups, elevated monocyte, neutrophil, and leukocyte counts in male ADHD participants may reflect subtle immune activation.

These sex-specific differences highlight the importance of considering biological sex in future studies of ADHD pathophysiology.

Given the predominantly null findings for other inflammatory markers and the cross-sectional nature of this study, these results should be interpreted with caution. These findings are best understood as hypothesis-generating, underscoring the need for longitudinal and mechanistic investigations. Future research incorporating neuroimaging, cytokine profiling, and multi-omics approaches may help to clarify the nature and clinical relevance of inflammation in ADHD. Such studies could also explore whether specific immune profiles might inform individualized treatment strategies for specific patient subsets.

Acknowledgements I thank Dr. Sertaç Çankaya for insightful discussions regarding statistical analyses and data interpretation.

The author presented portions of this research at the 37th European College of Neuropsychopharmacology (ECNP) Congress in Milan, Italy, September 21–24, 2024, where it received an 'ECNP Excellence Award' for poster presentations.

Author Contributions Conceptualization, Data curation, Formal analysis, Methodology, Writing – original draft & editing: Pınar Kızılay Çankaya**.**

Funding The author received no financial support for the research or publication of this article.

Data Availability The corresponding author can provide the dataset analyzed for this study upon request.

Declarations

Ethical Approval This retrospective study used data obtained for clinical purposes. This study was performed in accordance with the principles of the Declaration of Helsinki. Ethical approval was obtained from the Ethics Committee of the Koru Ankara Hospital on September 12, 2023 (approval number: 3201).

Informed Consent Not applicable.

Conflict of interest The author declares no conflict of interest related to this research.

References

1. American Psychiatric Association. Diagnostic and statistical manual of mental disorders. 5th ed. Washington, DC: American Psychiatric Association; 2013.
2. Faraone SV, Asherson P, Banaschewski T, Biederman J, Buitelaar JK, Ramos-Quiroga JA, Rohde LA. Attention-deficit/hyperactivity disorder. Nat Rev Dis Primers. 2015;1:15020. https://doi.org/10.1038/nrdp.2015.20.
3. Asherson P, Buitelaar J, Faraone SV, Rohde LA. Adult attention-deficit hyperactivity disorder: key conceptual issues. Lancet Psychiatry. 2016;3(6):568–78. https://doi.org/10.1016/S2215-0366(16)30032-3.
4. Caye A, Rocha TB, Anselmi L, Murray J, Menezes AM, Barros FC, Gonçalves H, Wehrmeister F, Jensen CM, Steinhausen HC, Swanson JM, Kieling C, Rohde LA. Attention-deficit/hyperactivity disorder trajectories from childhood to young adulthood: evidence from a birth cohort supporting a late-onset syndrome. JAMA Psychiatr. 2016;73(7):705–12. https://doi.org/10.1001/jamapsychiatry.2016.0383.
5. Polanczyk GV, Willcutt EG, Salum GA, Kieling C, Rohde LA. ADHD prevalence estimates across three decades: an updated systematic review and meta-regression analysis. Int J Epidemiol. 2014;43(2):434–42. https://doi.org/10.1093/ije/dyt261.
6. Stibbe T, Huang J, Becker M, Schupp J, Barke A, Klein EM, Kennes LN, Konrad K. Gender differences in adult ADHD: investigating cognitive control, emotion regulation, and their neural correlates. Front Psychiatry. 2020;11:562498. https://doi.org/10.3389/fpsyt.2020.562498.

7. Su W, Jia H, Yang L, Zhang J, Wei Z, Tsikwa P, et al. The etiology of attention deficit disorder with hyperactivity: a protocol for an umbrella review. PLoS One. 2025;20(1): e0318141. https://doi.org/10.1371/journal.pone.0318141.
8. Al-Gailani L, Al-Kaleel A. The relationship between prenatal, perinatal, and postnatal factors and ADHD: the role of nutrition, diet, and stress. Dev Psychobiol. 2024;66: e70004. https://doi.org/10.1002/dev.70004.
9. Luo Y, Weibman D, Halperin JM, Li X. A review of heterogeneity in attention deficit/hyperactivity disorder (ADHD). Front Hum Neurosci. 2019;13: 42. https://doi.org/10.3389/fnhum.2019.00042.
10. Takahashi N, Ishizuka K, Inada T. Peripheral biomarkers of attention-deficit hyperactivity disorder: current status and future perspective. J Psychiatr Res. 2021;137:465–70. https://doi.org/10.1016/j.jpsychires.2021.03.012.
11. Schnorr I, Siegl A, Luckhardt S, Wissmann M, Schröder S, Schulze MB, Hoffmann K, Spengler D, Bosch R, Faraone SV, Ramos-Quiroga JA, Aust G, Lesch KP. Inflammatory biotype of ADHD is linked to chronic stress: a data-driven analysis of the inflammatory proteome. Transl Psychiatry. 2024;14:37. https://doi.org/10.1038/s41398-023-02729-3.
12. Allred EN, Dammann O, Fichorova RN, Hooper SR, Hunter SJ, Joseph RM, Kuban K, Leviton A, O'Shea TM, Scott MN. Systemic inflammation during the first postnatal month and the risk of attention deficit hyperactivity disorder characteristics among 10 year-old children born extremely preterm. J Neuroimmune Pharmacol. 2017;12(3):531–43. https://doi.org/10.1007/s11481-017-9742-9.
13. Donfrancesco R, Nativio P, Di Benedetto A, Villa MP, Andriola E, Melegari MG, Cipriano E, Di Trani M. Anti-Yo antibodies in children with ADHD: first results about serum cytokines. J Atten Disord. 2020;24(11):1497–502. https://doi.org/10.1177/1087054716643387.
14. Anand D, Colpo GD, Zeni G, Zeni CP, Teixeira AL. Attention-deficit/hyperactivity disorder and inflammation: what does current knowledge tell us? A systematic review. Front Psychiatry. 2017;8: 228. https://doi.org/10.3389/fpsyt.2017.00228.
15. Kerekes N, Sanchéz-Pérez AM, Landry M. Neuroinflammation as a possible link between attention-deficit/hyperactivity disorder (ADHD) and pain. Med Hypotheses. 2021;157: 110717. https://doi.org/10.1016/j.mehy.2021.110717.
16. Sayed SZ, Hassan ZO, Abdelraheem WM, El-Shorbagy HH, Kamal NM. Is there a link between peripheral inflammation and blood brain barrier integrity in children with attention-deficit/hyperactivity disorder? A case-control study BMC Pediatr. 2024;24:769. https://doi.org/10.1186/s12887-024-05254-4.
17. Howren MB, Suls J, Lamkin DM. Associations of depression with C-reactive protein, IL-1, and IL-6: a meta-analysis. Psychosom Med. 2009;71(2):171–86. https://doi.org/10.1097/PSY.0b013e3181907c1b.
18. Miller BJ, Buckley P, Seabolt W, Mellor A, Kirkpatrick B. Meta-analysis of cytokine alterations in schizophrenia: clinical status and antipsychotic effects. Biol Psychiatry. 2011;70(7):663–71. https://doi.org/10.1016/j.biopsych.2011.04.013.
19. Hughes HK, Moreno RJ, Ashwood P. Innate immune dysfunction and neuroinflammation in autism spectrum disorder (ASD). Brain Behav Immun. 2022;108:245–54. https://doi.org/10.1016/j.bbi.2022.12.001.
20. Saccaro LF, Schilliger Z, Perroud N, Piguet C. Inflammation, anxiety, and stress in attention-deficit/hyperactivity disorder. Biomedicines. 2021;9(10):1313. https://doi.org/10.3390/biomedicines9101313.
21. Dunn GA, Nigg JT, Sullivan EL. Neuroinflammation as a risk factor for attention deficit hyperactivity disorder. Pharmacol Biochem Behav. 2019;182:22–34. https://doi.org/10.1016/j.pbb.2019.05.005.
22. Instanes JT, Halmøy A, Engeland A, Haavik J, Furu K, Klungsøyr K. Attention-deficit/hyperactivity disorder in offspring of mothers with inflammatory and immune system diseases. Biol Psychiatry. 2017;81(5):452–9. https://doi.org/10.1016/j.biopsych.2015.11.024.
23. Lin YT, Chen YC, Gau SF, Yeh TH, Fan HY, Hwang YY, Lee YL. Associations between allergic diseases and attention deficit hyperactivity/oppositional defiant disorders in children. Pediatr Res. 2016;80:480–5. https://doi.org/10.1038/pr.2016.111.
24. Lotter H, Altfeld M. Sex differences in immunity. Semin Immunopathol. 2019;41(2):133–5. https://doi.org/10.1007/s00281-018-00728-x.
25. Roved J, Westerdahl H, Hasselquist D. Sex differences in immune responses: hormonal effects, antagonistic selection, and evolutionary consequences. Horm Behav. 2017;88:95–105. https://doi.org/10.1016/j.yhbeh.2016.11.017.
26. Klein SL, Flanagan KL. Sex differences in immune responses. Nat Rev Immunol. 2016;16:626–38. https://doi.org/10.1038/nri.2016.90.
27. Holder MK, Blaustein JD. Puberty and adolescence as a time of vulnerability to stressors that alter neurobehavioral processes. Front Neuroendocrinol. 2014;35(1):89–110. https://doi.org/10.1016/j.yfrne.2013.10.004.

28. Toenders YJ, Laskaris L, Davey CG, Berk M, Milaneschi Y, Lamers F, Penninx BWJH, Schmaal L. Inflammation and depression in young people: a systematic review and proposed inflammatory pathways. Mol Psychiatry. 2022;27(1):315–27. https://doi.org/10.1038/s41380-021-01306-8.
29. Dunn SE, Perry WA, Klein SL. Mechanisms and consequences of sex differences in immune responses. Nat Rev Nephrol. 2024;20:37–55. https://doi.org/10.1038/s41581-023-00787-w.
30. Gutiérrez-Brito JA, Lomelí-Nieto JÁ, Muñoz-Valle JF, Oregon-Romero E, Corona-Angeles JA, Hernández-Bello J. Sex hormones and allergies: exploring the gender differences in immune responses. Front Allergy. 2025;5:1483919. https://doi.org/10.3389/falgy.2024.1483919.
31. Hegvik TA, Klungsøyr K, Instanes JT, Engeland A, Haavik J. Associations between attention-deficit/hyperactivity disorder and autoimmune diseases are modified by sex: a population-based cross-sectional study. Eur Child Adolesc Psychiatry. 2018;27(5):663–75. https://doi.org/10.1007/s00787-017-1056-1.
32. Babinski DE. Sex differences in ADHD: review and priorities for future research. Curr Psychiatry Rep. 2024;26(4):151–6. https://doi.org/10.1007/s11920-024-01492-6.
33. Zulfic Z, Weickert CS, Weickert TW, Liu D, Myles N, Galletly C. Neutrophil–lymphocyte ratio – a simple, accessible measure of inflammation, morbidity and prognosis in psychiatric disorders? Australas Psychiatry. 2020;28(4):454–8. https://doi.org/10.1177/1039856220908172.
34. Machingo TA, Bruton AM, Ryan JJ, Brock G, Bever AM, Liu C, McNamara RK. Neutrophil-lymphocyte ratio and platelet-lymphocyte ratio in ADHD: a systematic scoping review. Adv Neurodev Disord. 2024;8:233–46. https://doi.org/10.1007/s41252-024-00433-x.
35. Su M, Ouyang X, Song Y. Neutrophil to lymphocyte ratio, platelet to lymphocyte ratio, and monocyte to lymphocyte ratio in depression: a meta-analysis. J Affect Disord. 2022;308:375–83. https://doi.org/10.1016/j.jad.2022.04.038.
36. Mazza MG, Lucchi S, Tringali AGM, Rossetti A, Botti ER, Clerici M. Neutrophil/lymphocyte ratio and platelet/lymphocyte ratio in mood disorders: a meta-analysis. Prog Neuropsychopharmacol Biol Psychiatry. 2018;84(Pt A):229–36. https://doi.org/10.1016/j.pnpbp.2018.03.012.
37. Mazza MG, Lucchi S, Rossetti A, Clerici M. Neutrophil-lymphocyte ratio, monocyte-lymphocyte ratio and platelet-lymphocyte ratio in non-affective psychosis: a meta-analysis and systematic review. World J Biol Psychiatry. 2020;21:326–38. https://doi.org/10.1080/15622975.2019.1583371.
38. Cheng Y, Wang Y, Wang X, Jiang Z, Zhu L, Fang S. Neutrophil-to-lymphocyte ratio, platelet-to-lymphocyte ratio, and monocyte-to-lymphocyte ratio in depression: an updated systematic review and meta-analysis. Front Psychiatry. 2022;13: 893097. https://doi.org/10.3389/fpsyt.2022.893097.
39. Lin KB, Fan FH, Cai MQ, Yang Y, Mao SH, Wang ZW, Zhang LY, Zhang J, Shi Y, Cao FL. Systemic immune inflammation index and system inflammation response index are potential biomarkers of atrial fibrillation among the patients presenting with ischemic stroke. Eur J Med Res. 2022;27:106. https://doi.org/10.1186/s40001-022-00733-9.
40. Abus S, Kapici Y, Ayhan S, Guc B, Tekin A. Comparative study of frontal QRS-T angle and inflammatory parameters in obsessive compulsive disorder patients and healthy control group. Medicine (Baltimore). 2022;11(4):1473–7. https://doi.org/10.1097/MD.0000000000031479.
41. Öz E, Parlak ME, Kapıcı Y, Balatacı U, Küçükkelepçe O, Kurt F. Pre- and post-treatment evaluation of routine blood analysis in patients with attention deficit hyperactivity disorder and comparison with the healthy control group. Sci Rep. 2023;13:16437. https://doi.org/10.1038/s41598-023-43553-5.
42. Akıncı MA, Uzun N. Evaluation of hematological inflammatory markers in children and adolescents with attention deficit hyperactivity disorder. Bratisl Lek Listy. 2021;122(4):256–62. https://doi.org/10.4149/BLL_2021_042.
43. Avcil S. Evaluation of the neutrophil/lymphocyte ratio, platelet/lymphocyte ratio, and mean platelet volume as inflammatory markers in children with attention-deficit hyperactivity disorder. Psychiatry Clin Neurosci. 2018;72(7):522–30. https://doi.org/10.1111/pcn.12659.
44. Binici NC, Kutlu A. Is ADHD an inflammation-related disorder? Anadolu Psikiyatri Derg. 2019;20:313–20. https://doi.org/10.5455/apd.5132.
45. Elsadek AE, Al-Shokary AH, Abdelghani WE, Kamal NM, Ibrahim AO, El-Shorbagy HH, Suliman HA, Barseem NF, Abdel Maksoud YH, Azab SM, Nour El Din DM. Serum levels of interleukin-6 and tumor necrosis factor alpha in children with attention-deficit hyperactivity disorder. J Pediatr Neurosci. 2020;15:402–8. https://doi.org/10.4103/jpn.JPN_1_20
46. Önder A, Gizli Çoban Ö, Sürer Adanır A. Elevated neutrophil-to-lymphocyte ratio in children and adolescents with attention-deficit/hyperactivity disorder. Int J Psychiatry Clin Pract. 2021;25:43–8. https://doi.org/10.1080/13651501.2020.1804940.
47. Ceyhun HA, Gürbüzer N. New hematological parameters as inflammatory biomarkers: systemic immune inflammation index, platerethritis, and platelet distribution width in patients with adult attention deficit hyperactivity disorder. Adv Neurodev Disord. 2022;6(2):211–23. https://doi.org/10.1007/s41252-022-00258-6.

48. Abdel Samei AM, Mahmoud DAM, Salem Boshra B, Abd El Moneam MHE-D. The interplay between blood inflammatory markers, symptom domains, and severity of ADHD disorder in children. J Atten Disord. 2024;28(1):66–76. https://doi.org/10.1177/10870547231197213
49. Gędek A, Modrzejewski S, Gędek M, Wojtowicz K, Wojnarowski P, Utykański J, Wojnarowska M, Szubert P, Szrajda J, Liczbiński P. Neutrophil to lymphocyte ratio, platelet to lymphocyte ratio, and monocyte to lymphocyte ratio in ADHD: a systematic review and meta-analysis. Front Psychiatry. 2023;14:1258868. https://doi.org/10.3389/fpsyt.2023.1258868.
50. Garre JM, Yang G. Contributions of monocytes to nervous system disorders. J Clin Invest. 2018;128(10):4202–16. https://doi.org/10.1172/JCI99484.
51. Gibney SM, Drexhage HA. Evidence for a dysregulated immune system in the etiology of psychiatric disorders. J Neuroimmune Pharmacol. 2013;8(4):900–20. https://doi.org/10.1007/s11481-013-9462-8.
52. Misiak B, Wójta-Kempa M, Samochowiec J, Schiweck C, Aichholzer M, Reif A, Samochowiec A, Stańczykiewicz B. Peripheral blood inflammatory markers in patients with attention deficit/hyperactivity disorder (ADHD): a systematic review and meta-analysis. Prog Neuropsychopharmacol Biol Psychiatry. 2022;118: 110581. https://doi.org/10.1016/j.pnpbp.2022.110581.
53. Greven JK, Richards CU, Buitelaar JS. Sex differences in ADHD. In: Banaschewski T, Coghill D, Zuddas A, editors. Oxford textbook of attention deficit hyperactivity disorder. Oxford: Oxford Academic; 2018. p. 154–60.
54. Waddell J, McCarthy MM. Sexual differentiation of the brain and ADHD: what is a sex difference in prevalence telling us? Curr Top Behav Neurosci. 2012;9:341–60. https://doi.org/10.1007/7854_2011_169.
55. Leffa DT, Torres ILS, Rohde LA. A review on the role of inflammation in attention-deficit/hyperactivity disorder. Neuroimmunomodulation. 2018;25(5–6):328–33. https://doi.org/10.1159/000489635.
56. Yorbik O, Mutlu C, Koc D, Geyikçi İ. Platelet indices in children with attention deficit hyperactivity disorder. Bull Clin Psychopharmacol. 2014;24(3):231–7. https://doi.org/10.5455/bcp.20140623020856.
57. Zahorec R. Ratio of neutrophil to lymphocyte counts – rapid and simple parameter of systemic inflammation and stress in critically ill. Bratisl Lek Listy. 2001;102:5–14.
58. Yu H, Chen L, Liu Y, Li Z, Wang J, Li Y, Wu J, Liu P. Mean platelet volume as a biomarker in patients with major depressive disorder: a retrospective study. Front Psychiatry. 2020;11:573246. https://doi.org/10.3389/fpsyt.2020.573246.
59. Wysokiński A, Szczepocka E. Platelet parameters (PLT, MPV, P-LCR) in patients with major depressive disorder: a brief report. Neuropsychiatry (London). 2016;6(4):159–63. https://doi.org/10.4172/Neuropsychiatry.1000135.
60. Hariri M, Djazayery A, Djalali M, Saedisomeolia A, Rahimi A, Abdolahian E. Effect of n-3 supplementation on hyperactivity, oxidative stress and inflammatory mediators in children with attention-deficit-hyperactivity disorder. Malays J Nutr. 2012;18:329–35.
61. Gillies D, Leach MJ, Perez Algorta G. Polyunsaturated fatty acids (PUFA) for attention deficit hyperactivity disorder (ADHD) in children and adolescents. Cochrane Database Syst Rev. 2023;4: CD007986.
62. Miller BJ, Goldsmith DR. Towards an immunophenotype of schizophrenia: progress, potential mechanisms, and future directions. Neuropsychopharmacology. 2017;42(1):299–317. https://doi.org/10.1038/npp.2016.211.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

Pınar Kızılay Çankaya , is a psychiatrist in private practice in Ankara, Turkey, and a doctoral student in Medical Pharmacology at Gazi University. She completed her medical degree at Ege University (2008) and her psychiatry residency at Karadeniz Technical University (2015). Dr. Kızılay Çankaya received the ECNP Excellence Award in 2024 and has contributed to several peer-reviewed articles on topics including metabolic syndrome in psychiatric patients, consultation-liaison psychiatry, and non suicidal self-injury. She holds specialized training in cognitive behavioral therapy, dynamic psychotherapy, and sexual therapy. She is an active member of the Psychiatric Association of Turkey (PAT), European Psychiatric Association (EPA), and European College of Neuropsychopharmacology (ECNP), contributing to task forces on ADHD, mood disorders, and immunology in psychiatry.